

Suchandra Thapa

Software Developer

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PROFESSIONAL SUMMARY

Software engineer with 15+ years of experience building and operating backends for complex distributed environments. Experience building agentic systems using AWS Bedrock Agentcore and PydanticAI. Extensive work with Python, Go, C++, and Rust. Expertise with creating developer tooling and building DevOps/Platform engineering.

TECHNOLOGIES AND SKILLS

Programming Languages: Python, Go, C++, Rust, Scala, Java, Javascript

Cloud Platforms: AWS, Google Cloud Platform (GCP), Azure, Kubernetes, Docker

Databases: PostgreSQL, MySQL, MongoDB, Cassandra, Elasticsearch, Redis

Frameworks: Django, FastAPI, Flask, Vue

DevOps Tools: Terraform, FluxCD, ArgoCD, Github Actions, Docker, Helm, Git, Jenkins

Big Data: Flink, Hadoop, Spark, Airflow

Others: High Performance Computing (HPC), Security & Reliability Engineering, Distributed Data & Computing, Serverless, System Design, System Optimization

WORK EXPERIENCE

Software Developer/Platform Engineer
Fractional/Manifold

September 2025 - June 2026

Led investigations and implementation of features from custom code to using AWS bedrock agentcore. Investigated and created proof of concept using Agentcore Runtimes and Code Interpreter. Updated existing agents to use agentcore runtime features and implemented CI/CD pipelines to allow deployment of agentcore runtimes using github actions. Created terraform modules to allow Agentcore resources to be created and managed.

- Created MLOps and DevOps workflows using Terraform and AWS Bedrock Agentcore runtimes, memory, and Bedrock inference profiles
- Evaluated solutions and lead implementation of Go CLI tools to provide users with cross cloud file transfer capabilities (AWS S3, GCS)
- Created and deployed agents and agent orchestration to analyze life sciences data using PydanticAI and AWS Agentcore
- Migrated agentic sandboxes from custom lambda interface to Agentcore sandboxes to allow evaluation and execution of LLM generated code
- Created Terraform modules to create and manage AWS agent infrastructure
- Debugged issues in various PaaS workbenches based on customer bug reports
- Updated and implemented new features in various PaaS (RStudio, jupyterlab, positron) images to provide users

Freelance Software Developer
Self-Employed, Honolulu, HI

February 2024 – September 2025

Led major efforts to modernize existing website and infrastructure from using Django 1.10 and python 2.8 to Django 5.1 and python 3. Migrated existing bare-metal infrastructure to using containers running on AWS ECS and Fargate with Cloudwatch logging. Created online dashboards using R/Shiny to provide information on manufacturing data.

- Led effort to modernize existing website to a modern Django codebase
- Containerized backend services and moved existing services from bare metal to using AWS ECS and Fargate
- Created reporting infrastructure for manufacturing outputs using R/Shiny and SQLServer
- Added Cloudwatch logging and monitoring to backed services
- Active contributor to multiple open source projects including DocumentDB, MemXRay, and CPython

Wrote and maintained application that allows admins to deploy apps to several multi-tenant Kubernetes clusters simultaneously. Developed operators to enable multi-cloud services across multiple Kubernetes based clusters. Implemented CI/CD pipelines and GitOps using FluxCD to provided self service integration test beds for microservices. Instrumented services to provide OpenTelemetry observability and created infrastructure to collect and provide OpenTelemetry information.

- Created a Go Kubernetes operator that let a central service to administer and run services running on multiple Kubernetes clusters.
- Optimized multi-threaded C++ REST API server reducing memory utilization by 80%
- Added OpenTelemetry instrumentation, collectors, and dashboards for enhanced monitoring, performance tracking, and debugging to existing infrastructure (API servers, backend services, web servers)
- Improved developer experience significantly by writing GitOps pipelines that allowed developers to quickly instantiate distributed analysis services on Kubernetes clusters for testing and debugging
- Contributed ongoing maintenance and features to distributed analysis systems that allowed terabyte scale data analysis using clusters distributed around the world
- Spearheaded implementation of CI/CD pipelines using FluxCD and GitHub Actions to enhance release efficiency by automating software builds, testing, and deployments on Kubernetes clusters
- Optimized build and deployment pipelines from Github to Google Kubernetes Engine, reducing deployment times by 66+% to significant accelerate development cycles and testing feedback loops.

System Engineer
University of Chicago, Chicago, IL

November 2006 - October 2018

Architected and developed software used to support distributing computation and data across 100+ sites distributed internationally. Wrote code that supported computing infrastructure that provided over 1 million cpu hours daily and transfered over 1PB of data every week. Contributed to projects that discovered the Higgs boson, observed gravity waves and provided cutting edge results in the search for dark matter.

- Drove efforts to simplify integration of over 100 international computing sites into the Open Science Grid (OSG) infrastructure by architecting and developing software for configuring the OSG software stack.
- Facilitated groundbreaking brain research by over 20 research groups by creating a distributed computing service for processing and analyzing brain MRI scans
- Contributed to cutting edge research into dark matter by the Xenon1T collaboration by developing and optimizing Monte Carlo simulation workflows for dark matter research, scaling workloads to harness over 100,000 CPU hours per day across a distributed computing environments.
- Engineered a high-throughput, multi-petabyte data management system, ensuring secure and efficient data access across international research institutions.
- Created an analytics platform for monitoring job execution and data transfer across distributed computing environments, enhancing performance visibility and debugging capabilities.
- Collaborated with scientific researchers to scale complex computational workflows from single desktop environments to large-scale distributed infrastructures.
- Led integration and validation efforts for the full Open Science Grid (OSG) software stack, ensuring seamless deployment across research grid infrastructures.
- Optimized distributed data processing pipelines, improving research data accessibility and computational efficiency for various scientific projects.

EDUCATION

University of Chicago , Chicago IL
BA in Math / BA in Computer Science

OTHER ACTIVITIES

Help to organize and lead PyHawai'i, Honolulu Machine Learning, Code 4 Hawaii, and CloudHNL groups. Helped to organize UnVoxxed Hawaii, a unconference that featured some of the java developers in the world (GaalVM developers, Twitter JVM team, IBM JVM team, Azul JVM team). Active contributor to multiple open source projects including DocumentDB, MemXRay, and CPython.